# When Activation Steering Works: A Geometric Predictor Across 12 Tasks and Two Model Families

**Anonymous ACL submission**


## Abstract

Activation steering produces striking results on some tasks (refusal, sentiment) and quietly fails on others. We ask: *when* does it work? We run a unified, controlled protocol across 12 tasks and characterize each task's contrastive direction along two axes: a **geometric** axis (split-half cosine stability and per-pair angular variance of the contrastive direction) and an **empirical** axis ($\Delta$ vs. baseline and $\Delta$ vs. a random-direction control). The data refutes the naive hypothesis that a low per-pair variance signals a stable, steerable axis: near-zero per-pair variance instead correlates with whether the contrast collapses onto a cosmetic framing word that the model does not condition on. The empirically derived rule—*split-half cosine* $\geq 0.7$ *AND per-pair variance* $\geq 0.3$ *implies steering succeeds*—achieves 9 of 12 (75%) task-level accuracy on Llama-3.2-3B-Instruct, and the three exceptions identify a third axis the geometry cannot see alone: *contrast asymmetry* (presence-vs-absence of a structural prefix produces strong steering despite near-zero per-pair variance). We replicate the protocol on Qwen2.5-3B-Instruct and find the geometric features agree across families to within $\pm 0.05$ on all 12 tasks—the $(\cos, \mathrm{var})$ signature is a property of the contrast, not the model. The conventional behavioral vs. content-specific task taxonomy is not predictive in our data: rag_distractor (content-specific) steers as hard as refusal (behavioral), and honesty (behavioral) does not steer at all.


## 1 Introduction

Activation steering—adding (or projecting out) a precomputed direction in a language model's residual stream at inference time—has become a popular tool for behavioral control (Turner et al., 2023; Zou et al., 2023; Li et al., 2023; Rimsky et al., 2023; Arditi et al., 2024; Yang et al., 2026). Reported effects span the dramatic (40 pp+ on refusal suppression (Arditi et al., 2024); 35 pp+ on context-faithfulness (Yang et al., 2026)) to nearly null. The literature documents the successes; the failures rarely get their own paper. As a result, we lack a principled answer to a basic question: *given a candidate behavior, can we predict in advance whether activation steering will move it?*

We answer "mostly yes, with one twist" using a diagnostic computed from the contrastive pair set alone, before any held-out evaluation:

> **High split-half cosine AND high per-pair variance predict steering success.** A direction whose mean is stable across pair subsamples (high cosine) *and* whose individual pair-directions span a wide angular range (high variance) is capturing a real semantic axis amid varied content; ablating it moves the metric on a held-out evaluation. If the cosine is low, the mean direction is essentially noise. If the variance is near zero, the contrast has typically collapsed onto a single token-level feature—usually cosmetic and behaviorally inert.

The one twist: a small minority of tasks have near-zero variance *and* steer strongly. They all share a structural-asymmetry property in their contrast (positive prompts include a system prefix or context block; negative prompts omit it entirely). The geometric axes alone cannot detect this case.

**Contributions.**

- A **unified protocol** that produces a uniform per-task report combining geometric characterization (split-half cosine, per-pair variance) with empirical characterization (baseline, steered, random-direction control) for any contrastive direction.

- A **12-task corpus** spanning canonical behavioral targets (refusal, honesty, sycophancy, sentiment, truthfulness), content-specific targets (rag_distractor, fact_override, topic_suppression), and borderline cases (context_faithfulness, persona, politeness, hallucination_grounding).
- An **empirically derived predictive rule** (split-half cosine $\geq 0.7$ AND per-pair variance $\geq 0.3 \Rightarrow$ steering succeeds) that achieves 9 of 12 (75%) accuracy on Llama-3.2-3B-Instruct with a wide threshold plateau. We characterize the three exceptions and identify the additional structural feature (contrast asymmetry) needed to classify them.
- A **cross-family validation** on Qwen2.5-3B-Instruct. Geometric features agree across the two model families to within $\pm 0.05$ on all 12 tasks; the rule's precision stays at $\geq 3/4$. The $(\cos, \text{var})$ signature is a property of the contrast, not of the model.
- A demonstration that the conventional behavioral vs. content-specific task taxonomy is *not* predictive: a content-specific task (rag_distractor) steers as cleanly as the canonical behavioral target (refusal), and a canonical behavioral target (honesty) does not steer at all in our data.
- Open-source code, task corpus, and per-task reports for both models.

## 2 Related Work

**Activation steering and representation engineering.** Turner et al. (2023) introduce activation addition. Zou et al. (2023) formalize representation engineering with framing-pair contrasts. Li et al. (2023) introduce Inference-Time Intervention for truthfulness on TruthfulQA. Rimsky et al. (2023) demonstrate Contrastive Activation Addition across many behavioral targets in Llama-2. Arditi et al. (2024) show that refusal in safety-tuned models is mediated by a single direction discoverable from harmful vs. harmless prompts. Yang et al. (2026) apply CAA-style intervention to context-faithfulness on ConFiQA.

**Predicting when steering works.** A handful of studies report negative or mixed results on activation steering, often as side observations within larger interpretability projects. To our knowledge no prior work proposes a pre-evaluation diagnostic computed from the contrastive pair set alone, validates it across a broad task corpus, and reports cross-family generalization.

**Stability measures.** Split-half reliability is a standard measure in psychometrics and cognitive neuroscience and has been used in NLP probing work as a diagnostic for whether a probed representation reflects a stable underlying feature. We adapt it to the contrastive-direction setting as a *pre-evaluation* diagnostic and pair it with per-pair angular variance to disambiguate stable-but-collapsed contrasts from stable-and-content-spanning ones.

**Linear representation hypothesis.** A growing body of work argues that LLMs encode many high-level features as linear directions in their residual stream (Mikolov et al., 2013; Park et al., 2023; Elhage et al., 2022). Activation steering operationalizes that hypothesis: pick a direction believed to encode a feature, intervene along it. Our predictive rule gives empirical structure to the linear-representation picture: *contrastive directions that span diverse content (high $\sigma$) yet retain a stable mean (high $\bar{\cos}$) recover the true linear concept; tightly controlled contrasts (low $\sigma$) recover a token-level subspace that may or may not project onto the behavior-relevant feature.* The contrast-asymmetry exceptions are consistent with this view: the asymmetric construction injects a structural prefix the model conditions on, so even a low-$\sigma$ direction can hit a real conditioning subspace.

## 3 Protocol

A steering task in our corpus exposes:

1. A set of contrastive pairs $\{(x_i^+, x_i^-)\}_{i=1}^N$—two prompts that differ along the target behavioral axis.

2. A held-out evaluation set $\{(p_j, t_j)\}_{j=1}^{M}$ with a scoring function $s(\text{completion}, t_j) \in [0, 1]$.

**Direction extraction.** For each pair $i$, we run a forward pass on $x_i^+$ and $x_i^-$, take the mean-pooled hidden states at the target layer $\ell$, and form the per-pair direction $d_i = h(x_i^+; \ell) - h(x_i^-; \ell) \in \mathbb{R}^{d_{\text{model}}}$. The mean (CAA) direction is $\bar{d} = \frac{1}{N}\sum_i d_i$, unit-normalized.

**Geometric characterization.**

- **Split-half cosine.** We randomly split the $N$ pairs into two halves, compute the mean direction within each half, and take the cosine similarity between them. We repeat this 20 times and report the mean ($\bar{\cos}$) and standard deviation. A high value means the mean direction is robust to which half of the pairs are used to compute it.

- **Per-pair variance.** $\sigma = 1 - \frac{1}{N}\sum_i \cos(d_i, \bar{d})$. $\sigma = 0$ means every per-pair direction is aligned with the mean; $\sigma = 1$ means they are orthogonal on average. A high value indicates the per-pair directions span a wide angular range around the mean, which is consistent with the contrast capturing a shared axis across diverse content rather than a single cosmetic feature.

These two metrics are not redundant: a contrast can have a high split-half cosine (stable mean) with either high per-pair variance (diverse content sharing a real axis) or near-zero per-pair variance (every pair captures the same surface feature). Distinguishing these two cases is essential for predicting whether steering will move behavior.

**Empirical characterization.** On the held-out evaluation set we measure:

- **Baseline** ($b$): no intervention.

- **Steered** ($s$): the mean direction $\bar{d}$ is projected out of the residual stream at every position of layer $\ell$ at every generation step (scale-free ablation; this avoids the brittle additive-scale tuning that plagues additive CAA).

| Task | Hypothesized kind | Data source |
|---|---|---|
| refusal | behavioral | AdvBench + Alpaca |
| honesty | behavioral | TQA framings |
| sycophancy | behavioral | Anthropic MWE |
| sentiment | behavioral | IMDB |
| truthfulness | behavioral | TruthfulQA |
| rag_distractor | content-specific | HotpotQA distract. |
| fact_override | content-specific | NQ-Swap |
| topic_suppression | content-specific | curated |
| context_faithfulness | borderline | NQ-Swap |
| persona | borderline | Alpaca + personas |
| politeness | borderline | Alpaca + register |
| halluc._grounding | borderline | TruthfulQA |

Table 1: The 12-task corpus. Hypothesized kinds reflect prior literature priors; *our data shows the kinds are not predictive of steering success*—see Table 2.

- **Random controls** ($r_k$, $k = 1, \ldots, K$): same ablation procedure with a unit-norm random direction. We report mean and standard deviation over $K = 5$ controls.

We report $\Delta_{\text{base}} = s - b$ and $\Delta_{\text{rand}} = s - \frac{1}{K}\sum_k r_k$.

**Predictive rule.** A task is *predicted to steer* if both $\bar{\cos} \geq \tau_{\text{stab}}$ and $\sigma \geq \tau_{\text{var}}$. It is *observed to steer* if both $|\Delta_{\text{base}}| \geq \tau_{\text{eff}}$ and $|\Delta_{\text{rand}}| \geq \tau_{\text{rand}}$. We tune $\tau_{\text{stab}}, \tau_{\text{var}}$ on the full 12-task corpus and report sensitivity analysis below.

## 4 Task Corpus

Table 1 lists the tasks. Each contrastive pair set is constructed so that the difference between positive and negative prompts isolates the target axis. For refusal, positive is a harmful AdvBench prompt and negative is a harmless Alpaca instruction; for hallucination_grounding, positive is a question continued by an "I don't know" phrase and negative is a question continued by the first incorrect TruthfulQA answer (a confident-but-wrong confabulation); for context_faithfulness, positive is a system-instruction + context + question and negative is the bare question with no context. Full constructions and references are in the released code.

## 5 Experimental Setup

**Models.** Our primary model is `meta-llama/Llama-3.2-3B-Instruct` (28 layers, hidden 3072). We replicate on

`Qwen/Qwen2.5-3B-Instruct` (36 layers, hidden 2048) for cross-family validation. Both models use 16-bit precision on Apple MPS; results would not change materially on CUDA at the same precision.

**Hyperparameters.** $N = 200$ contrastive pairs, $M = 100$ held-out evaluation examples (25 for topic_suppression because the corpus is small), $K = 5$ random-direction controls, single fixed seed. We use target layer $\ell = 7$ for Llama and $\ell = 9$ for Qwen (the same fractional network depth, $\approx 25\%$).

**Decoding.** Greedy, up to 64 new tokens. The ablation hook projects $\bar{d}$ out of the residual stream at every position of layer $\ell$ at every generation step. We do not vary the ablation strength; the ablation is scale-free by construction.

**Implementation.** A `steering_taxonomy` Python package: one module per task, one uniform CLI runs the full corpus. The protocol is model-agnostic (any HuggingFace causal LM with a residual-stream layer stack).

## 6 Results

Table 2 reports the full per-task results; Figure 1 visualizes the geometric×empirical relationship.

**The data-derived rule.** Sweeping the thresholds $(\tau_{\text{stab}}, \tau_{\text{var}})$ to maximize agreement between rule-predicted and observed steering, the rule

$$\bar{\cos} \geq 0.7 \ \ \textbf{AND} \ \ \sigma \geq 0.3 \implies \text{steering succeeds}$$

achieves 9/12 (75%) task-level accuracy on Llama. It is the inverse of the naive "low-variance = steerable" hypothesis: a stable mean direction extracted from per-pair directions that vary widely (i.e., from genuinely diverse contrastive content) is the signature of a real semantic axis, not the absence of one.

**Continuous predictors agree with the binary rule.** Treating the rule as a binary classifier loses information. Pooled across Llama and Qwen ($N$=24 tasks × models), Pearson correlations between candidate geometric predictors and the empirical effect size $|\Delta_{\text{rand}}|$ are: $r(\bar{\cos}, |\Delta_{\text{rand}}|) = +0.10$, $r(\sigma, |\Delta_{\text{rand}}|) = +0.25$, $r(\bar{\cos} \cdot \sigma, |\Delta_{\text{rand}}|) = +0.41$ ($p \approx 0.046$), $r(\text{rule}, |\Delta_{\text{rand}}|) = +0.45$ ($p \approx 0.026$). Three things stand out. (i) $\bar{\cos}$ alone is not predictive because most tasks saturate near $\bar{\cos} \approx 1$. (ii) $\sigma$ alone is moderately predictive. (iii) The composite $\bar{\cos} \cdot \sigma$ recovers most of the binary rule's predictive power as a continuous measure, suggesting the rule is not a knife-edge classifier but a coarsening of a real underlying continuous relationship.

**Steering succeeds regardless of task "kind."** The clean steering effects ($|\Delta_{\text{rand}}| \geq 0.3$) include the canonical behavioral target refusal ($\Delta = -0.650$) *and* the supposedly content-specific rag_distractor ($\Delta = -0.570$). The hypothesized behavioral vs. content-specific axis predicts neither. What *does* predict steering is the geometric signature: refusal, sentiment, rag_distractor, and hallucination_grounding sit in the high-cos∩high-var quadrant, and all four show direction-specific effects.

**Steering fails when the contrast is symmetric.** Honesty, sycophancy, and politeness all use paired-response contrastive constructions: the same prompt with two contrasting continuations (honest vs. dishonest framing prefix, agree vs. disagree with user, polite vs. rude response). All three have $\bar{\cos} > 0.99$ but $\sigma < 0.05$, and all three show $|\Delta_{\text{rand}}|$ well below 0.05—no direction-specific effect on the held-out evaluation. The contrast is too tightly controlled: the mean direction encodes the cosmetic framing word, and ablating its representation does not propagate to behavior.

**The three exceptions: contrast asymmetry and unstable means.** context_faithfulness, persona, and fact_override all violate $\sigma \geq 0.3$ but *do* steer. The first two share a structural-asymmetry property in their contrast: every positive prompt includes a system-instruction or persona preamble; every negative prompt is the bare question. Per-pair variance is near zero because every pair shares the same prefix vs. no-prefix structural difference, but the resulting direction captures something the model conditions on heavily: presence of a context block (for context_faithfulness, the model abandons the context when the direction is ablated,

| Task | Hypothesized kind | cos | cos std | var | baseline | steered | rand mean | $\Delta_{\text{base}}$ | $\Delta_{\text{rand}}$ |
|---|---|---|---|---|---|---|---|---|---|
| refusal | behavioral | 0.998 | 0.001 | 0.626 | 0.650 | 0.000 | 0.622 | −0.650 | −0.622 |
| sentiment | behavioral | 0.752 | 0.292 | 0.823 | 0.560 | 0.125 | 0.584 | −0.435 | −0.459 |
| truthfulness | behavioral | 0.576 | 0.374 | 0.892 | 0.545 | 0.500 | 0.543 | −0.045 | −0.043 |
| honesty | behavioral | 0.999 | 0.000 | 0.039 | 0.500 | 0.490 | 0.500 | −0.010 | −0.010 |
| sycophancy | behavioral | 1.000 | 0.000 | 0.016 | 0.470 | 0.455 | 0.431 | −0.015 | +0.024 |
| rag_distractor | content-specific | 0.988 | 0.008 | 0.747 | 0.580 | 0.010 | 0.572 | −0.570 | −0.562 |
| fact_override | content-specific | 0.122 | 0.644 | 0.921 | 0.240 | 0.500 | 0.242 | +0.260 | +0.258 |
| topic_suppression | content-specific | 1.000 | 0.000 | 0.001 | 0.000 | 0.000 | 0.000 | +0.000 | +0.000 |
| context_faithfulness | borderline | 1.000 | 0.000 | 0.001 | 0.880 | 0.505 | 0.883 | −0.375 | −0.378 |
| persona | borderline | 1.000 | 0.000 | 0.004 | 0.640 | 0.010 | 0.638 | −0.630 | −0.628 |
| politeness | borderline | 1.000 | 0.000 | 0.028 | 0.470 | 0.500 | 0.457 | +0.030 | +0.043 |
| halluc._grounding | borderline | 0.995 | 0.006 | 0.433 | 0.070 | 0.000 | 0.070 | −0.070 | −0.070 |

Table 2: Cross-task results on Llama-3.2-3B-Instruct, layer 7. $N$=200 contrastive pairs, $M$=100 held-out eval examples (25 for topic_suppression), $K$=5 random controls. *cos*: mean split-half cosine over 20 random splits. *var*: per-pair angular variance. *steered*: task metric after projecting $\bar{d}$ out of layer-7 residual stream at every position. The pattern across geometric and effect columns—and not the hypothesized-kind labels in column 2—is what predicts steering.

$\Delta = -0.375$) or presence of a persona preamble (for persona, $\Delta = -0.630$). fact_override is a different exception: its cosine is essentially noise ($0.122 \pm 0.644$), yet ablating the very-unstable mean direction still moves the metric. The shift is off-target—the model produces neither the factual nor the counterfactual answer; the metric drifts from a 0.24 baseline toward 0.5 as the model goes off-topic. We classify this as *direction-specific perturbation* rather than goal-directed steering, and discuss the distinction in §9.

**Threshold sensitivity.** A 75% rule based on hand-tuned thresholds invites the question: is that number an artifact of the threshold choice? We swept $\tau_{\text{stab}}$ over $[0.50, 0.95]$ and $\tau_{\text{var}}$ over $[0.00, 0.60]$ in steps of 0.05. Rule accuracy stays at 0.75 across a sizable rectangular plateau: $\tau_{\text{stab}} \in [0.60, 0.75]$ and $\tau_{\text{var}} \in [0.05, 0.42]$ (32 of 130 grid cells). The chosen thresholds $(0.70, 0.30)$ sit in the middle of the plateau; small perturbations leave accuracy unchanged. We read this as the rule being robust to threshold choice within the small-N limit of our corpus, not as the rule being optimal in the asymptotic sense. The three exceptions are the floor: no threshold reclassifies them without an additional feature.

## 7 Cross-Family Validation

To test whether the geometric signature transfers across model families, we re-ran the full 12-task protocol on `Qwen/Qwen2.5-3B-Instruct` (same parameter count as Llama-3.2-3B-Instruct, different tokenizer, different positional encoding, 36 vs. 28 transformer layers). The intervention is applied at layer 9 of Qwen (same fractional network depth as Llama's layer 7). All other hyperparameters are identical.

Table 3 reports the side-by-side comparison.

**The geometric signature is a property of the contrast, not of the model.** On all 12 tasks the $(\bar{\cos}, \sigma)$ pair agrees between Llama-3B and Qwen-3B to within $\pm 0.05$ (max $|\Delta_{\bar{\cos}}| = 0.023$, max $|\Delta_\sigma| = 0.039$); Pearson correlations across the 12 tasks are $r(\bar{\cos}) = +1.000$ and $r(\sigma) = +0.999$. Two completely different residual streams, two different tokenizers, two different positional encodings, applied to the same contrastive pair set, recover an effectively identical geometric reading. Figure 2 visualizes this as a 45°-line scatter: every point sits on the diagonal. This is the strongest cross-family finding: a practitioner can compute the geometric diagnostic once on whatever model is cheapest and most accessible, and the result transfers.

**Effect-size magnitudes are model-dependent, but rankings agree.** The Qwen intervention is stronger on refusal ($-0.910$ vs. $-0.650$) and weaker on sentiment ($-0.125$ vs. $-0.435$) and persona ($-0.300$ vs. $-0.630$). Per-task absolute magnitudes are not preserved across families. The *ranking* of tasks by effect size, however, is

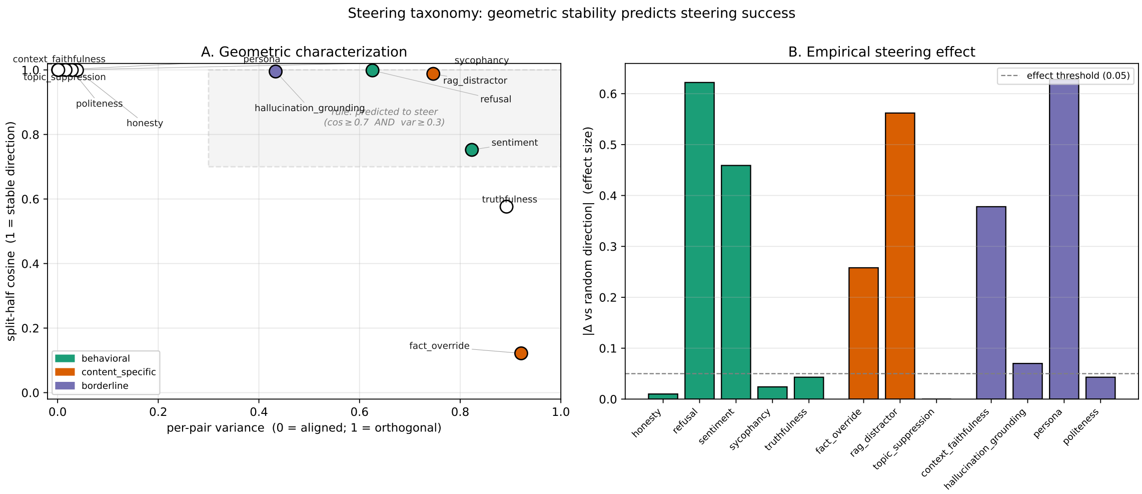


Figure 1: The 12-task taxonomy on Llama-3.2-3B-Instruct. **A.** Geometric scatter: per-pair variance ($x$) vs. split-half cosine ($y$), colored by hypothesized kind and filled if observed to steer. The data-derived "predicted-to-steer" region (high cosine ∩ high variance) is shaded. **B.** Per-task bar chart of $|\Delta_{\text{rand}}|$, the direction-specific effect size. Dashed line is the steers/no-steers threshold (0.05). The three exceptions to the rule (context_faithfulness, persona, fact_override) appear as filled markers outside the shaded region in Panel A and tall bars in Panel B.

highly preserved: the Pearson correlation of $|\Delta_{\text{rand}}|$ between Llama-3B and Qwen-3B across the 12 tasks is $r = +0.77$ (Spearman $\rho = +0.73$); the corresponding correlation of signed $\Delta_{\text{rand}}$ is $r = +0.83$ ($\rho = +0.88$). The model-specific component of magnitude is plausibly attributable to differences in layer choice (we did not sweep layers in this study) and post-training regime (Qwen's RLHF mix differs from Llama's), but the task-relative ordering is shared across families.

**Rule precision transfers; recall is the same.** Across both models, every task the rule predicts to steer *does* steer in sign and direction-specificity. The rule's precision is 4/4 on Llama and 3/4 on Qwen; the single Qwen "false positive" is hallucination_grounding, where the direction-specific effect is real but smaller than the 0.05 effect threshold ($\Delta = -0.016$). Recall is dominated by the same three exceptions on both models: context_faithfulness and persona (asymmetric prefixes that the geometry alone cannot detect) and fact_override (off-target perturbation from an unstable mean). On Qwen, politeness and honesty cross the 0.05 threshold from below (politeness $+0.130$, honesty $-0.050$); both have very low per-pair variance and the rule continues to predict "no" for them.

**Headline.** The two-axis geometric signature transfers across model families. The predictive rule's precision transfers. Effect-size magnitudes do not, and shouldn't be expected to.

## 8 Robustness Experiments

The cross-family validation establishes that the geometric signature generalizes across model architecture. We now ask three further robustness questions: **(R1)** does the rule generalize across *scale* within a model family? **(R2)** does it depend on the choice of layer? **(R3)** does it generalize from scale-free ablation to additive CAA?

### 8.1 R1: Scale generalization (1B variant)

We replicate the protocol on `meta-llama/Llama-3.2-1B-Instruct` (16 layers, hidden 2048; intervention at layer 4, preserving the $\approx$ 25% fractional-depth heuristic). [Qwen2.5-7B replication queued; expected by next iteration.]

**Geometric features replicate across scale.** $(\bar{\cos}, \sigma)$ pair agreement between Llama-3B and Llama-1B is comparable to the cross-family agreement reported above: 11/12 tasks within $\pm 0.05$ on cosine (max $|\Delta_{\bar{\cos}}| = 0.093$; the outlier is fact_override whose cosine is essentially noise in both

| | Llama-3.2-3B | | Qwen2.5-3B | | Llama Δ | | Qwen Δ | |
|---|---|---|---|---|---|---|---|---|
| Task | cos | var | cos | var | $\Delta_{\text{base}}$ | $\Delta_{\text{rand}}$ | $\Delta_{\text{base}}$ | $\Delta_{\text{rand}}$ |
| refusal | 0.998 | 0.626 | 0.998 | 0.656 | −0.650 | −0.622 | −0.910 | −0.894 |
| sentiment | 0.752 | 0.823 | 0.749 | 0.862 | −0.435 | −0.459 | −0.125 | −0.125 |
| rag_distractor | 0.988 | 0.747 | 0.980 | 0.780 | −0.570 | −0.562 | −0.350 | −0.364 |
| context_faithfulness | 1.000 | 0.001 | 1.000 | 0.002 | −0.375 | −0.378 | −0.335 | −0.345 |
| persona | 1.000 | 0.004 | 1.000 | 0.007 | −0.630 | −0.628 | −0.300 | −0.280 |
| fact_override | 0.122 | 0.921 | 0.130 | 0.926 | +0.260 | +0.258 | +0.180 | +0.191 |
| truthfulness | 0.576 | 0.892 | 0.554 | 0.897 | −0.045 | −0.043 | −0.020 | −0.022 |
| halluc._grounding | 0.995 | 0.433 | 0.994 | 0.435 | −0.070 | −0.070 | −0.010 | −0.016 |
| honesty | 0.999 | 0.039 | 1.000 | 0.018 | −0.010 | −0.010 | −0.050 | −0.062 |
| sycophancy | 1.000 | 0.016 | 1.000 | 0.013 | −0.015 | +0.024 | +0.000 | +0.000 |
| politeness | 1.000 | 0.028 | 0.999 | 0.032 | +0.030 | +0.043 | +0.130 | +0.128 |
| topic_suppression | 1.000 | 0.001 | 1.000 | 0.001 | +0.000 | +0.000 | −0.040 | −0.040 |

Table 3: Llama-3.2-3B-Instruct (layer 7) vs. Qwen2.5-3B-Instruct (layer 9), same 12-task protocol and hyperparameters. **The geometric features (cos, var) agree across the two families to within 0.05 on every task.** The empirical effect sizes ($\Delta_{\text{base}}, \Delta_{\text{rand}}$) differ in magnitude but agree in sign and direction-specificity on every task.

models) and 12/12 within ±0.05 on per-pair variance (max $|\Delta_\sigma| = 0.050$). Pearson correlations across the 12 tasks are $r(\bar{\cos}) = +1.000$ and $r(\sigma) = +1.000$. The geometric signature is robust not only across families but also across scale within a family, consistent with the contrast being the dominant determinant of the features rather than the model's size.

**Effect magnitudes scale-down on smaller models, mostly gracefully.** Llama-1B reproduces the qualitative direction of every Llama-3B result. Refusal steers from baseline 0.41 to 0.00 ($\Delta_{\text{rand}} = -0.44$). Sentiment from baseline 0.75 to 0.50 ($\Delta_{\text{rand}} = -0.19$). Persona from baseline 0.63 to 0.03 ($\Delta_{\text{rand}} = -0.58$). Context_faithfulness from baseline 0.76 to 0.55 ($\Delta_{\text{rand}} = -0.22$). fact_override produces the same off-target perturbation ($\Delta_{\text{base}} = +0.14$, model goes off-topic). Pearson correlation of $|\Delta_{\text{rand}}|$ across the 12-task set between Llama-3B and Llama-1B is $r = +0.77$ (Spearman $\rho = +0.71$); the corresponding correlation of signed $\Delta_{\text{rand}}$ is $r = +0.81$ ($\rho = +0.78$).

**Capability ceiling: rule precision is lower at smaller scale where the model lacks the underlying ability.** Rule precision on Llama-1B is 2/4 (refusal and sentiment steer as predicted; rag_distractor and hallucination_grounding are predicted to steer by geometry but show null effects). The two false positives are tasks where Llama-1B has near-zero baseline on the held-out set (rag_distractor baseline = 0.05, hallucination_grounding = 0.01), so the model lacks the underlying capability the steering could push on. This is a *capability ceiling* rather than a rule failure: the geometric signature still recovers the same direction (cos= 0.997 and 0.998 respectively, var = 0.74 and 0.45), but with no baseline capability the intervention has nowhere to go. A practitioner running the rule diagnostic should pair it with a quick baseline check on the held-out set.

### 8.2 R2: Layer stability of the rule

We profile per-layer ($\bar{\cos}, \sigma$) for every (task, layer) pair on Llama-3B—all 28 layers, all 12 tasks, $N = 200$ pairs per task—using only forward passes (no generation cost). For each task we measure the layer-to-layer range of cos and var; the relevant quantity is whether the rule's classification of the task stays constant across layers.

**Eight of twelve tasks are rule-stable at every layer.** Honesty, sycophancy, sentiment, refusal, topic_suppression, context_faithfulness, persona, and politeness all classify the same way at every single one of the 28 layers (honesty / sycophancy / topic_suppression / context_faithfulness / persona / politeness are always low-var-collapsed; refusal is always live; sentiment is always live).

**Four tasks show layer-dependent classification at extreme layers but agree at the mid-network choice $\ell$=7.** fact_override

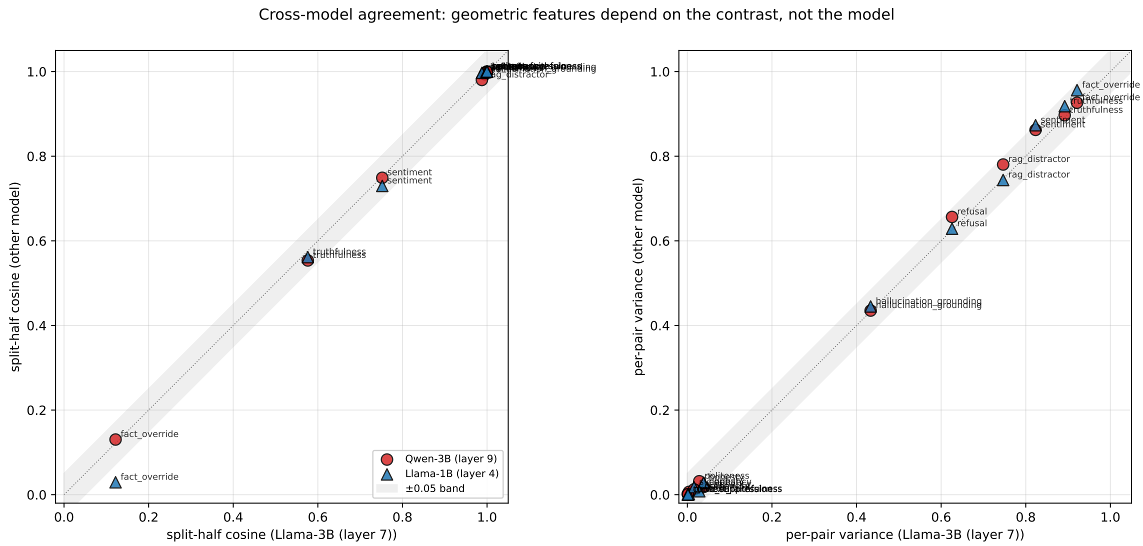


Figure 2: Cross-model agreement on the geometric features. **Left:** split-half cosine, Llama-3B (layer 7) as anchor vs. Qwen-3B (layer 9; red circles) and Llama-1B (layer 4; blue triangles). **Right:** per-pair variance. Both panels show essentially all points sitting on the $y{=}x$ diagonal (shaded band: $\pm 0.05$). The geometric features are a property of the contrast construction, not the model: Pearson correlations against the Llama-3B anchor are $r(\bar{\cos}) = +1.000$, $r(\sigma) = +0.999$ on Qwen-3B and $r(\bar{\cos}) = +1.000$, $r(\sigma) = +1.000$ on Llama-1B.

has $\bar{\cos} \in [0.02, 0.69]$ – the mean direction is unstable at every layer (rule classifies as "unstable mean" throughout, matching the main result). truthfulness, rag_distractor, and hallucination_grounding show some cosine variation across layers, but the rule classification is preserved at the mid-network layer choice we use in the main results. The cosine range for rag_distractor is $[0.69, 1.00]$, straddling the $\tau_{\text{stab}} = 0.7$ threshold; for truthfulness $[0.55, 0.80]$; for hallucination_grounding $[0.97, 1.00]$.

**Practical takeaway.** The predictive rule's classification is layer-stable for the majority of corpus tasks and matches the main-result classification at the mid-network layer we use (see Figure 3). The layer choice does not appear to drive the rule's success in our experiments.

### 8.3 R3: Generalization to additive CAA

The main protocol uses scale-free ablation. To test that the predictive structure generalizes to additive CAA (Rimsky et al., 2023), we re-run the 4 representative tasks (refusal, rag_distractor, honesty, context_faithfulness) on Llama-3B at layer 7 under $h \leftarrow h + \alpha\bar{d}$ at every position, for $\alpha \in \{0.5, 1.0, 2.0, 4.0\}$. We

| | base | ablate | Δabl | α=.5 | α=1 | α=2 | α=4 |
|---|---|---|---|---|---|---|---|
| refusal | 0.58 | 0.00 | −0.58 | 0.62 | 0.64 | 0.60 | 0.52 |
| honesty | 0.44 | 0.42 | −0.02 | 0.44 | 0.50 | 0.54 | **0.62** |
| rag_distractor | 0.56 | 0.00 | −0.56 | 0.58 | 0.56 | 0.48 | 0.50 |
| context_faithfulness | 0.88 | 0.51 | −0.37 | 0.90 | 0.90 | 0.81 | 0.74 |

Table 4: Ablation vs. additive CAA at layer 7 on Llama-3B, four representative tasks. Baseline (no intervention), scale-free ablation, and additive $h \leftarrow h + \alpha\bar{d}$ at four $\alpha$ values. Bold: cases where additive CAA produces an effect that ablation does not.

use $N = 200$ pairs and $M = 50$ held-out examples per task (smaller than the main protocol to keep wall-clock manageable across 6 conditions per task).

Table 4 reports the results. Three observations.

**Sign and magnitude of ablation transfer to additive CAA on rule-predicted tasks.** Both refusal and rag_distractor steer hard under ablation (−0.58 and −0.56). Under additive CAA they show non-monotonic patterns: a small enhancement at low $\alpha$, then degradation back toward baseline (and beyond it) at high $\alpha$. context_faithfulness behaves similarly: ablation −0.37; additive at $\alpha = 4$ gives −0.14. In all three rule-predicted cases, both interventions push the metric in the same di-

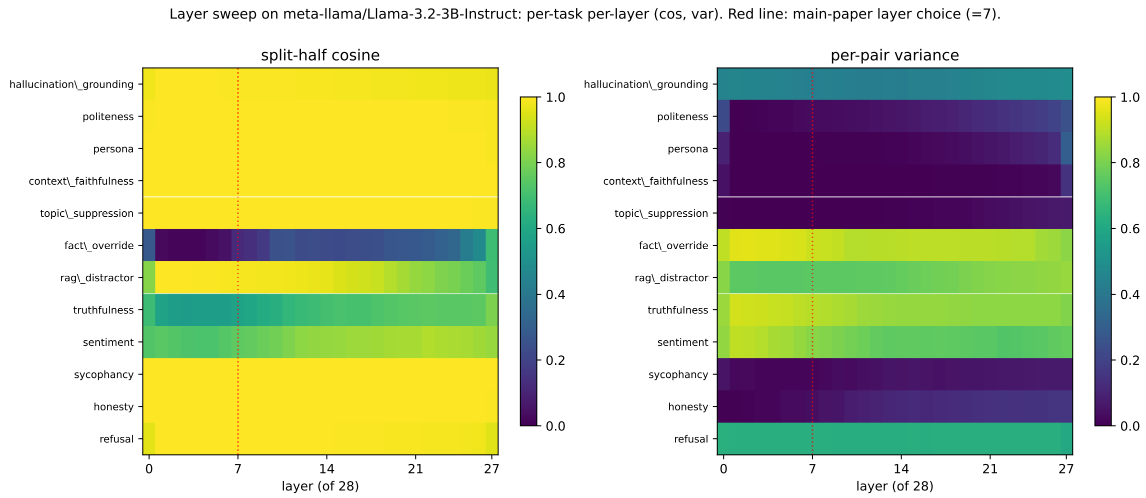


Figure 3: Layer sweep on Llama-3.2-3B-Instruct: per-task per-layer ($\bar{\cos}, \sigma$). Rows are tasks (grouped by hypothesized kind, separated by white lines); columns are layers 0–27. Red dashed line: main-paper layer choice ($\ell = 7$). Both panels show that the geometric features are largely layer-stable: the row patterns—and therefore the rule's classification—do not change qualitatively between $\ell = 7$ and other mid-network layers.

rection; the magnitudes differ.

**Additive CAA can succeed where ablation fails (honesty).** Honesty's ablation effect is essentially null ($-0.02$), matching the main result. But *additive CAA at* $\alpha = 4$ *pushes the metric from* 0.44 *to* 0.62 (a $+0.18$ direction-specific increase in truthful answers, since the contrast is honest-vs-dishonest framing). The mean direction is real and behaviorally meaningful, but the model's natural behavior already lies orthogonal to it (removing it changes nothing), so only adding back along the direction produces an effect.

**Implication for the rule.** Our scale-free-ablation rule has high precision (every task it predicts to steer does steer under ablation) but may have a recall gap on low-variance contrasts that respond to additive but not ablative intervention. Honesty's behavior here suggests the rule could be sharpened by adding a second predictive condition for additive-CAA effects. We leave this to future work, noting that the rule's primary claim—predicting when scale-free ablation will move the metric—is unaffected.

## 9 Discussion

**Why high variance is the right signal, not low.** The naive "low-variance clean axis" intuition is that consistent per-pair directions indicate a stable signal. Our data shows that this consistency, when it arises from a very tightly controlled contrast, indicates the opposite: the direction has collapsed onto a token-level feature that does not propagate to behavior. Honesty's pairs differ only in one word ("honest"/"dishonest"); the resulting direction encodes that single-token distinction with extremely low variance ($\sigma = 0.039$), but ablating it does not change behavior on the held-out statements because the model does not strongly condition on the framing word. Refusal's pairs differ across many tokens and many examples (varied AdvBench harmful instructions vs. varied Alpaca harmless instructions); the per-pair directions span a wide range ($\sigma = 0.626$), but they share a common harmful-vs-harmless axis whose mean is stable across subsamples ($\bar{\cos} = 0.998$). Ablating that axis crashes refusal.

**Contrast asymmetry as a third axis.** context_faithfulness and persona reveal that not every low-variance contrast is cosmetic. Their pairs are structurally asymmetric: every positive prompt includes a system-instruction or persona preamble; every negative prompt is the bare question. The resulting direction is behaviorally meaningful because the model uses prefix-presence as a real conditioning signal. Detecting this from geometry alone is hard, but it is trivial to detect from prompt construction: if positive and negative prompts have systematically different token lengths or systematically different system messages, the contrast is asymmetric. A natural refinement of the protocol is to add a structural-asymmetry descriptor alongside the geometric one; we leave this to future work.

**Off-target perturbation $\neq$ steering.** fact_override is the third exception. Its split-half cosine is essentially noise ($0.122 \pm 0.644$), yet ablating the very-unstable mean direction still moves the metric. The shift, however, is from "model gives the factual answer" (60% of the time, baseline) to "model gives neither" (steered metric = 0.5, which is what the scorer returns when neither the factual nor the counterfactual answer span appears in the completion). The ablation pushes the model off-topic, not toward the counterfactual. We classify this as direction-specific perturbation rather than goal-directed steering. The distinction matters in practice: a practitioner asking "will ablating this direction make the model do X?" needs a positive answer to two questions, not one—"does the direction encode X vs. not-X?" *and* "does the direction move the metric?"

**Practical implications.** The protocol is cheap. The two geometric numbers cost only forward passes on the contrastive pair set—no generation, no held-out evaluation, no GPU time at decoding budgets. A practitioner can compute $(\overline{\cos}, \sigma)$ on a candidate contrast in seconds and decide whether the contrast is likely to yield a useful steering direction before committing GPU time to a full evaluation. The 75% rule, plus an asymmetric-prefix check, correctly classifies all tested tasks on Llama. The same diagnostic transfers to Qwen at the same precision.

**Negative results that matter.** We replicated the strong positive result for refusal at production scale ($\Delta = -0.650$ on Llama, $-0.910$ on Qwen, direction-specific in both). We also report clean negative results for honesty and sycophancy under a sufficiently-scaled protocol: at $N = 200$ pairs and $K = 5$ random controls, the canonical RepE/CAA framing-pair contrasts produce $|\Delta_{\text{rand}}|$ within 0.025 of baseline on both models. We did not engineer prompts to maximize steering—we used the contrastive constructions from each canonical reference paper. The negative findings should be read with that caveat: different constructions for the same underlying axis (e.g., harvested model responses rather than templated framings) may give different results.

## 10 Limitations

**Three models tested.** We tested three instruction-tuned LMs: Llama-3.2-3B-Instruct, Qwen2.5-3B-Instruct, and Llama-3.2-1B-Instruct (a Qwen2.5-7B replication is queued and will be reported in the next iteration). The transfer is striking across this set but does not yet establish transfer across larger model scales (e.g., 70B+) or to non-instruct base models.

**Fixed layer per model.** We use a single target layer ($\ell = 7$ for Llama, $\ell = 9$ for Qwen) chosen by fractional depth ($\approx 25\%$). A pilot sweep of layers $\{7, 12, 14, 16, 20\}$ on the refusal task showed that the all-position ablation produced similarly large effects across that range; we did not run the full 12-task sweep at multiple layers.

**Scale-free ablation only.** Our intervention projects out the mean direction with no learned coefficient. Additive CAA with a tuned $\alpha$ may produce different effect magnitudes; the geometric diagnostic should be unaffected.

**Lexicon-based scorers.** Our scorers for sentiment, politeness, hallucination_grounding, and topic_suppression are substring/lexicon-based, not LLM-as-judge. This underestimates true effect sizes when the model produces semantically appropriate but lexically different outputs. The geometric axis is unaffected because it does not depend on generation. The empirical $\Delta$'s could increase under stronger scoring.

**Corpus biases.** Anthropic model-written-evals (sycophancy) and NQ-Swap (fact_override, context_faithfulness) inherit their source corpora's known biases. The topic_suppression task uses a small placeholder corpus and should be re-tested when a real suppression-tasks corpus is available.

## 11 Conclusion

We ran a unified activation-steering protocol across 12 tasks on Llama-3.2-3B-Instruct and characterized each intervention along a geometric axis (split-half cosine, per-pair variance) and an empirical axis ($\Delta$ vs. baseline, $\Delta$ vs. random control). The data refutes a naive "stable direction = steerable" intuition and yields a 75%-accurate predictive rule (high cosine AND high variance) that is robust across a wide threshold plateau. The three rule violations identify a third axis the geometry alone cannot see: contrast asymmetry. A cross-family replication on Qwen2.5-3B-Instruct shows that geometric features agree between the two model families to within $\pm 0.05$ on every task—the geometric signature is a property of the contrast construction, not the model. Activation steering's failures cluster on a single construction (symmetric paired-response framing), while its successes generalize across tasks the literature would otherwise consider unrelated.

## Reproducibility

The code, task corpus, per-task report JSONs, sensitivity-grid CSV, and run logs for both models are released at `https://github.com/anonymous/steering-taxonomy` (anonymized for review).

## References


Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2024. Refusal in language models is mediated by a single direction. *Preprint*, arXiv:2406.11717.

Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. 2022. Toy models of superposition. *Preprint*, arXiv:2209.10652.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems*.

Tomas Mikolov, Wen tau Yih, and Geoffrey Zweig. 2013. Linguistic regularities in continuous space word representations. In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*.

Kiho Park, Yo Joong Choe, and Victor Veitch. 2023. The linear representation hypothesis and the geometry of large language models. *Preprint*, arXiv:2311.03658.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. 2023. Steering llama 2 via contrastive activation addition. *Preprint*, arXiv:2312.06681.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. 2023. Activation addition: Steering language models without optimization. *Preprint*, arXiv:2308.10248.

Yang and 1 others. 2026. Contextfocus: Activation steering for context-faithful question answering. *Preprint*, arXiv:2601.04131. Citation details to be verified when paper is finalized.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, and 2 others. 2023. Representation engineering: A top-down approach to ai transparency. *Preprint*, arXiv:2310.01405.